

College of Engineering and Computing

SWEN 360 – Software Design & Engineering

Assignment 01

Requirements Analysis and Initial Design

BatchBite
Preorder and Production Platform for Small Food Businesses

Team Information

Team Names: Shaikha, Nehad, Aaya, Zain, Malaak, Marwa

Team Members & Roles:

- Scrum Master: Shaikha
- Development Team Members: Nehad, Aaya, Zain, Malaak, Marwa

A. Team Deliverables

Sprint Planning

Sprint Duration: October 1 – October 18

Sprint Goal: Sprint 1 aims to lay the groundwork for the development of the BatchBite system through setting up the functional and non-functional requirements of the system, creating the user stories for the entire group of users, as well as the creation of the design artifacts of the system, such as the wireframes, UML diagrams, and the architecture of the system. In addition, the GitHub repository will be set up.

Task Breakdown:

- Functional Requirements: Malaak
- Non-Functional Requirements: Aaya
- User Stories (Customer & Administrator): Aaya
- User Stories (Business Owner & Kitchen Staff): Malaak
- Wireframes: Marwa
- UML Diagrams (Use Case, Class, Sequence): Nehad & Zain
- Activity Diagrams: Marwa
- System Architecture Diagram: Nehad & Zain
- Documentation & Meeting Logs: Shaikha
- GitHub Repository & Kanban Board Management: Shaikha
- Sprint Planning & Task Tracking: Shaikha
- Sprint Retrospective: Shaikha & All Team Members

Stand-ups Summary

All meetings were approximately 15–20 minutes. A total of 8 meetings were held/planned for Sprint 1.

Date	Day	Attendance	Start Time	End Time	Notes
1/10	Thursday	6/6	12:30	12:45	Discussed the idea behind BatchBite and how it could help small food businesses manage preorders and organize food preparation. Identified the main features and outlined the initial sprint objectives.
4/10	Sunday	4/6 (Nehad and Marwa attended Online)	12:40	1:00	Distributed tasks among team members based on their assigned roles. Agreed on the main deliverables and discussed how to divide the requirements, user stories, and design work.
6/10	Tuesday	5/6 (Zain Was Not Present)	12:30	12:50	Organized the project repository on GitHub and created folders for the team's documents and design files. Reviewed the task board and discussed how to keep track of individual responsibilities.
8/10	Thursday	6/6 (Online)	5:00	5:20	Discussed the expected system requirements and the different people who would use BatchBite. Began outlining user needs and reviewing the features required for each type of account.

13/10	Tuesday	3/6 (Nehad, Marwa and Zain were not present)	12:30	12:50	Reviewed progress on the interface designs and system diagrams. Team members discussed their assigned design tasks, shared updates, and identified the remaining work needed to bring the initial designs together.
15/10	Thursday	6/6	12:30	12:50	Compared the progress of the different project components and discussed whether the requirements and user stories aligned with the planned system features. Reviewed outstanding tasks and updated their progress on the board.
16/10	Friday	6/6 (Online)	4:30	4:50	Focused on bringing the initial deliverables together. Reviewed the documentation and design files, checked the organization of the GitHub repository, and discussed corrections needed before completing the assignment.
18/10	Sunday	6/6 (Online)	7:00	7:30	Reviewed the sprint's overall progress and the status of the assigned deliverables. Discussed difficulties encountered during the work, considered improvements for future teamwork, and reviewed the remaining submission requirements.

Sprint Retrospective

Strengths:

- Maintained regular communication through the eight stand-up meetings to share updates and coordinate progress.
- Divided responsibilities among team members, allowing everyone to focus on their assigned deliverables.
- Used the GitHub repository and Kanban board to organize project files, assign tasks, and monitor progress.
- Collaborated on the requirements, user stories, wireframes, and UML diagrams to establish the foundation for BatchBite.

Challenges:

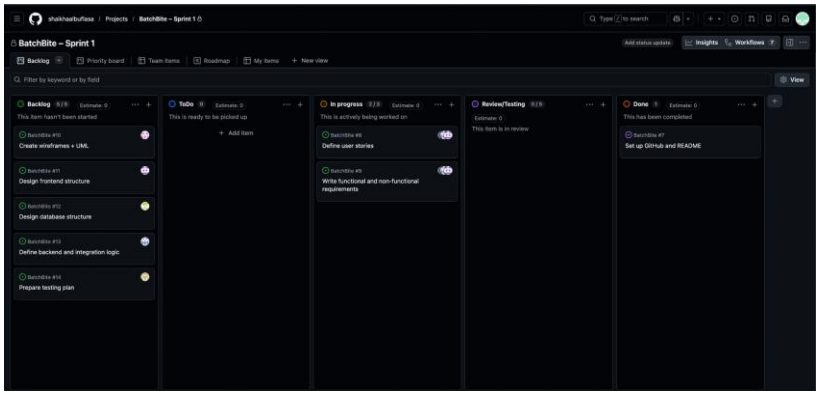
- Some team members were absent from meetings, which occasionally affected coordination and progress.
- Overlapping responsibilities in UI/UX design and UML diagrams required additional communication to avoid duplicated work.
- The team needed time to become familiar with GitHub and organize commits and documentation effectively.
- Limited time reflection made it difficult to discuss every challenge and evaluate the team's performance in depth.

Improvements for Next Sprint:

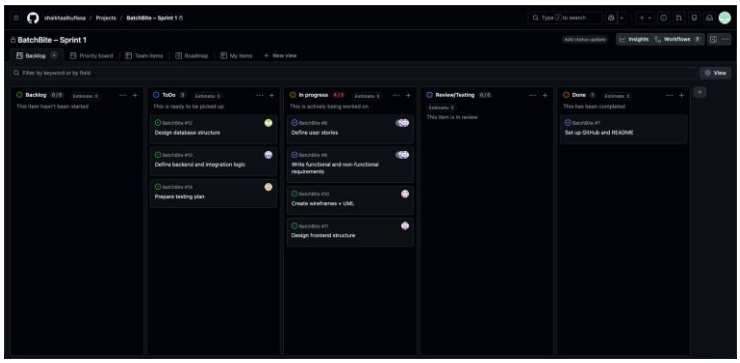
- Establish a consistent meeting schedule and encourage all team members to attend regularly.
- Define task ownership and deliverables more clearly to reduce overlapping work.
- Improve GitHub practices by committing changes regularly and linking relevant commits to tasks on the project board.
- Allocate more time for the sprint retrospective to discuss what worked well, what needs improvement, and how the team can work more efficiently.
- Update the Kanban board consistently so that task statuses accurately reflect the team's progress.

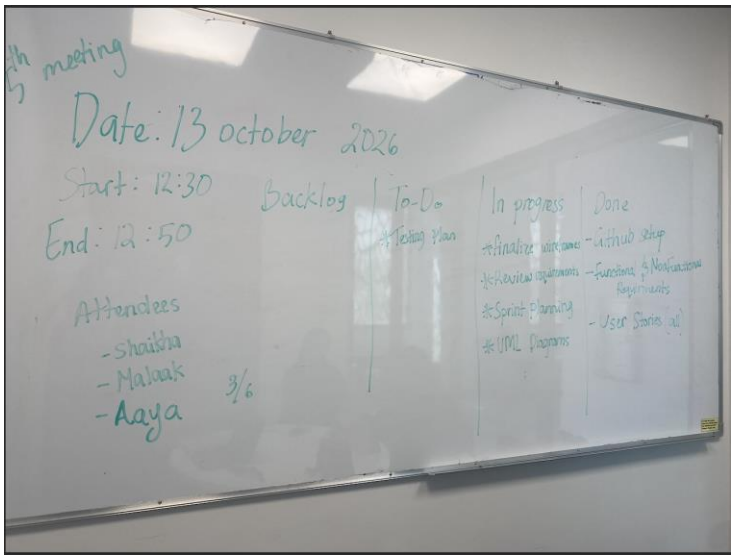
Kanban Board Progression

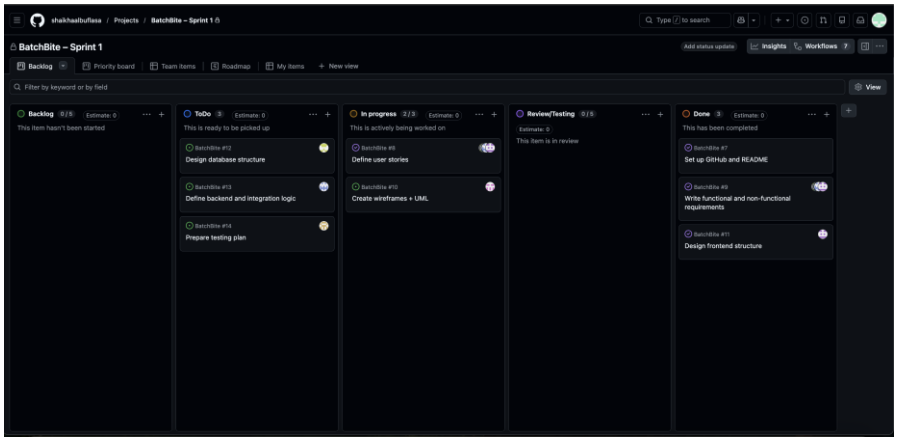
Beginning (October 6): Initial state of the BatchBite Kanban board on October 6, 2026, with tasks organized by status to track the team's progress throughout Sprint 1.



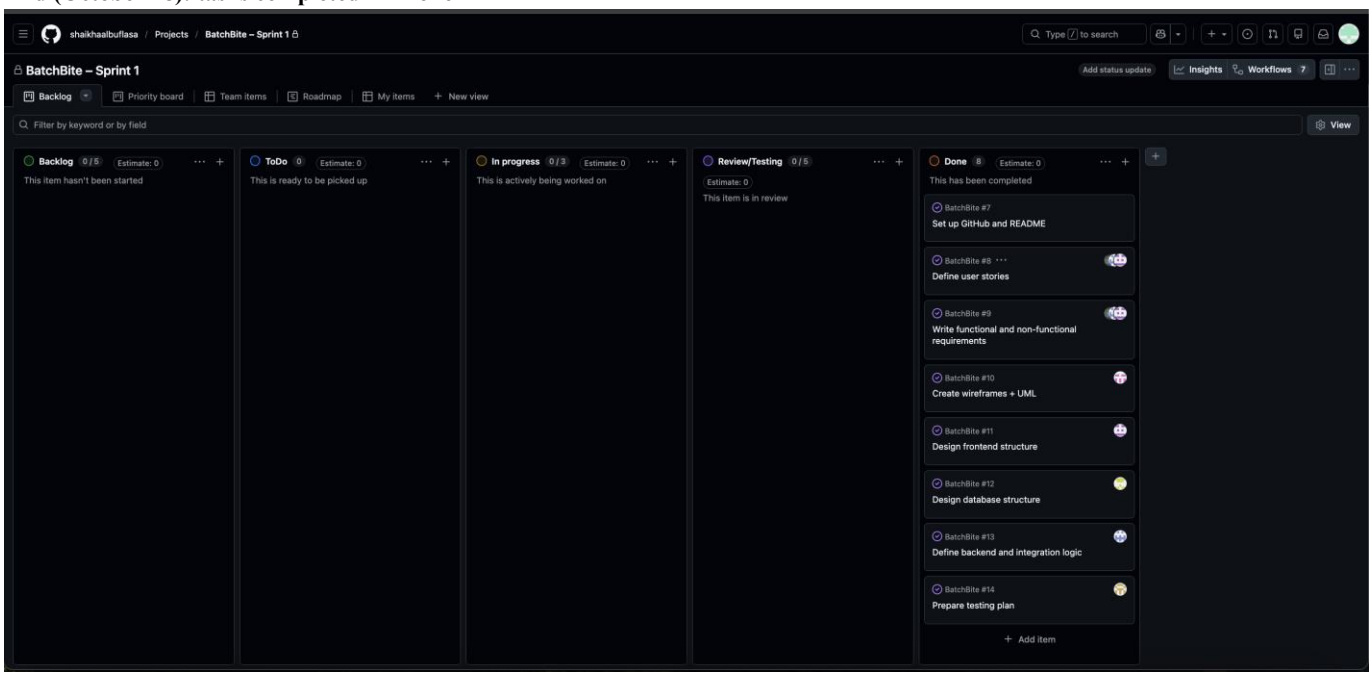
Mid-Sprint (October 8): Multiple tasks in progress



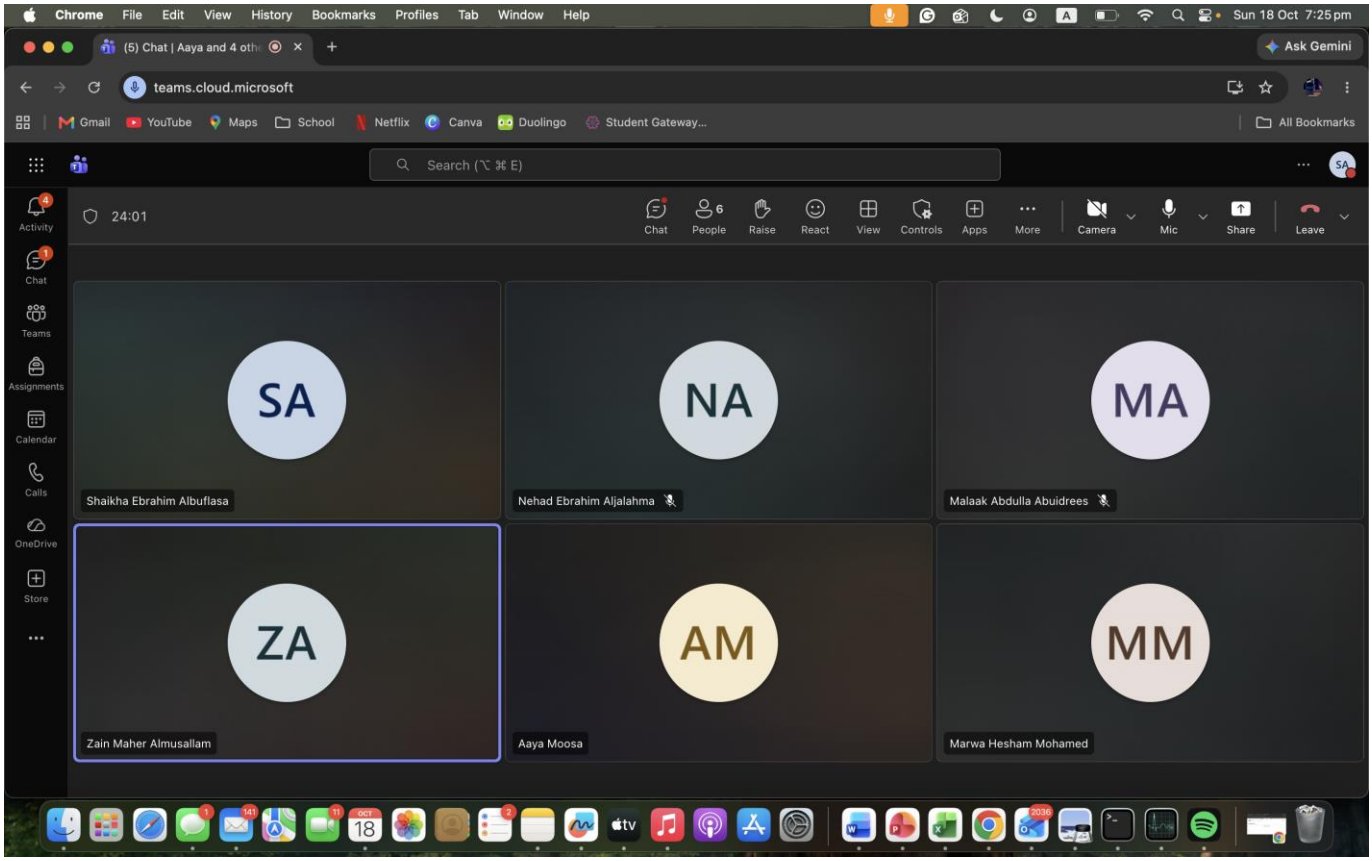




End (October 18): tasks completed in 'Done'



https://github.com/shaikhaalbuflasa/BatchBite



October 18: A meeting was conducted online to finalize sprint 1 requirements

2. Initial Product Artifacts

Functional Requirements

This section describes actions that BatchBite shall do. The requirements are categorized based on modules and are assigned to a priority level (High, Medium, and Low), along with the sprint during which they will be implemented. Requirements for Sprint 1 make up the primary core requirements; others are added now to incorporate them into the design.

1. **Functional Requirements**

This section describes actions that BatchBite shall do. The requirements are categorized based on modules and are assigned a priority level (High, Medium, and Low) along with the sprint during which they will be implemented. Requirements for Sprint 1 make up the primary core requirements; others are added now to incorporate them into the design.

1.1 Accounts and Access

ID	Requirement	Priority	Sprint
FR-1	The system will provide role-based account options for Customers, Business Owners, Kitchen Staff, and Administrators.	High	1
FR-2	The system will provide options for secure registration and login, and will limit access to all screens and functions based on the role permitted to access them.	High	1
FR-3	The system will provide Administrators with the ability to manage user accounts and set or modify user roles.	Medium	1

1.2 Products and Production Batches

ID	Requirement	Priority	Sprint
FR-4	The system shall provide Business Owners with the ability to create and modify listings of their products with pictures, descriptions, variants, ingredients information, pricing, and available quantity.	High	1
FR-5	The system shall provide Business Owners with the ability to create batches for product manufacturing with an order deadline, maximum quantity, and pick-up date.	High	1
FR-6	The system shall automatically close orders for a particular batch upon reaching the deadline or maximum quantity.	High	1

1.3 Cart, Checkout, and Stock Reservation

ID	Requirement	Priority	Sprint
FR-7	The system shall allow Customers to browse open batches and place products in their cart.	High	1
FR-8	The system shall show the order summary (Items, quantities, total, pickup slot) before the customer confirms the checkout.	High	1
FR-9	The system shall confirm a payment through a simulated payment confirmation.	High	1
FR-10	The system shall reserve the stock when the checkout occurs by holding it, in a way that the stock will not be accessible to another Customer while the hold is active	High	1
FR-11	The system shall cancel a reservation of the stock and the capacity when a hold expires or when an order is cancelled.	High	1

1.4 Pickup Slots

ID	Requirement	Priority	Sprint
FR-12	The system shall enable the Business Owners to set pickup timeslots with maximum orders per time slot.	High	1
FR-13	The system shall enable Customers to choose a suitable pickup time slot and prevent them from selecting time slots that are full.	High	1

1.5 Order Tracking and Production Workflow

ID	Requirement	Priority	Sprint

FR-14	The system shall monitor each order throughout the process of Confirmed, Preparing, Ready, and Collected.	High	1
FR-15	The system shall only permit valid status changes (from Confirmed to Preparing to Ready to Collected) and will prevent invalid status changes such as marking an order as Collected before its status becomes Ready.	High	1
FR-16	The system shall enable customers to access their order history and status of each order.	High	1
FR-17	The system shall provide a Business Owner Dashboard to view orders, cancellation of orders, and progress of production.	High	1
FR-18	The system shall enable the Kitchen Staff to access orders for the current batch along with customization and packaging information.	High	1
FR-19	The system shall remove canceled orders from the active production board.	Medium	1
FR-20	The system shall inform Customers about order confirmation, order ready for pick up, and delay notifications and maintain a record of all the notifications made accessible to the staff.	Medium	1
FR-21	The system shall consider two successive actions of a particular action as one action and shall not create duplicate records.	Medium	1

1.6 Order Changes and Cancellations

ID	Requirement	Priority	Sprint
FR-22	The system shall allow the Customers to modify quantity/variant prior to the cutoff provided only if there is sufficient stock and production capacity, recalculate the totals and the simulated refund but retain the same history of the order.	Medium	2
FR-23	The system shall maintain the recipe version and price version that were used for each confirmed order to ensure that future edits cannot affect the already confirmed orders.	Medium	2

1.7 Ingredients, Collection, and Reporting

ID	Requirement	Priority	Sprint
FR-24	The system shall derive the required ingredients based on confirmed orders, scaling the recipes with the quantity of the order, compare them against available and reserved stock, and create a purchasing list for any shortfall.	Medium	2
FR-25	The system shall create a unique QR code for each order and only confirm the delivery of the order when it is Ready.	Medium	2

FR-26	The system shall organize the products into preparation batches on the production board retaining all customer-specific notes.	Low	2
FR-27	The system shall report the amount of orders confirmed/cancelled/produced/collected by batch and product.	Low	3

Non-Functional Requirements

This section discusses the quality attributes and constraints that BatchBite must satisfy to offer a reliable, secure and user-friendly experience to its customers, business owners, kitchen staff and administrators.

ID	Non-Functional Requirement	Priority
NFR-1	**Usability:** The system shall offer a clear and consistent interface that allows users to explore the features offered depending on their roles.	High
NFR-2	**Performance:** The system shall display product lists, order details, and order status information in not more than 3 seconds under normal operating conditions. .	High
NFR-3	**Security:** The system shall restrict access to certain pages based on user roles and authentication is needed for protected functions.	High
NFR-4	**Data Integrity:** The system shall ensure accurate product quantities, production capacity, pickup-slot availability, and order records when placing or canceling orders.	High
NFR-5	**Reliability:** The system shall prevent invalid order-status transition and duplicate orders as a result of multiple submissions.	High
NFR-6	**Compatibility:** The application shall offer a consistent experience when accessed using popular web browsers and mobile platforms.	Medium
NFR-7	**Maintainability:** The system shall have a modular architecture, and the functionalities can be changed independently without necessarily impacting other functions such as the product list, checkout and order tracking.	Medium
NFR-8	**Availability of Information:** Customers shall be able to access their order status and order history while authorized staff shall be able to access relevant information required for their roles.	High

These requirements will guide the design and implementation of BatchBite throughout the development sprints.

User Story: Customer

. User Stories: Customer

The Customer uses BatchBite to browse available products, place and manage orders, select pickup times, and stay informed about order progress.

ID	User Story	Acceptance Criteria	Related FR	Priority	Sprint
CU-1	As a Customer, I want to browse through available products and batches of production in order to see the available food products and select the one I want to order.	The system displays the available products along with their pictures, description, variants, price, and available quantity. The customer can see open batches and cannot make an order in closed batches.	FR-4, FR-5, FR-6, FR-7	High	1
CU-2	As a Customer, I want to add items to my cart and see my order summary, so that I can check my selected products, quantities, total, and pickup time slot before placing the order.	Customers can add products to the cart and set their quantities. The system shows the selected products, quantities, total and pickup time slot before placing the order.	FR-7, FR-8	High	1
CU-3	As a Customer, I would like to place an order and get payment confirmation, so that I will know that my order has been successfully placed.	The system reserves the required stock during placing an order and shows the simulated payment confirmation. The order is confirmed when the checkout process is completed. In case the checkout fails or there is no required stock, then no order is created.	FR-9, FR-10, FR-11	High	1
CU-4	As a Customer, I would like to select an available pickup time slot, so that I can pick up my order in a convenient time period.	The system shows available pickup time slots for the selected batch. Customers can choose time slots that have available capacity and cannot choose the ones that are full. The selected time slot is added to the order summary before confirming the checkout.	FR-8, FR-12, FR-13	High	1
CU-5	As a Customer, I would like to track the status of my order and see my order history, so that I will know the information about my orders and previous purchases.	The system shows the customer's orders and their statuses. Order statuses are following each other according to the order: Confirmed, Preparing, Ready, Collected. Customers can access their previous orders and see the latest status of each order.	FR-14, FR-15, FR-16	High	1
CU-6	As a Customer, I would like to get notifications about my orders, so that I will know when my order	The system sends the customers notifications when their orders are confirmed and ready to pick up. Also, when the order is delayed,	FR-20	Medium	1

	is confirmed, ready to pick up, or delayed.	the system sends the delay notification. The system saves all notifications, which the authorized staff can access.			
CU-7	As a Customer, I would like to update my order before the cutoff time, so that I can change my quantities and product variants if I need.	Customers can update their quantities and product variants before the cutoff time. The system checks the availability of stock and capacity before accepting the update and calculating the new order total. When the total changes, the system updates the simulated refund accordingly if necessary. The system saves the history of updates.	FR-22	Medium	2
CU-8	As a Customer, I would like to cancel my order when cancellation is allowed, so that I can cancel order I no longer need.	When cancellation is allowed, the system registers the cancellation, frees the reserved stock and capacity, and cancels the order from the active production board. The cancellation is reflected in the order records	FR-11, FR-17, FR-19	Medium	1

User Story: Business Owner

4. User Stories: Business Owner

The Business Owner is the owner of a small food business who schedules production batches, creates/manages orders, and tracks their progress. The stories below include acceptance criteria and functional requirements.

ID	User Story	Acceptance Criteria	Related FR	Priority	Sprint
BO-1	As a Business Owner, I would like to be able to create product listings with photos, variants, ingredients and pricing, in order to present the exact offerings to the customers.	Product save occurs for all fields and the product appears in the customer catalogue; errors appear for missing required fields.	FR-4	High	1
BO-2	As a Business Owner, I would like to be able to create production batches with deadlines, max quantities and collection dates, in order to avoid overselling.	Batch creation includes all three values; orders close automatically either at the set deadline or when the maximum quantity is reached.	FR-5, FR-6	High	1

BO-3	As a Business Owner, I would like to be able to create pickup slots with order limits, in order to avoid overcrowding during collection periods.	When the slot is full, the field is indicated as such and cannot be chosen by customers.	FR-12, FR-13	High	1
BO-4	As a Business Owner, I would like to be able to see all the orders in a particular batch in a dashboard, in order to get my confirmed demand.	Dashboard displays every order with information on customer, ordered items, current status and assigned pick-up slot along with the total amount of ordered products compared to maximum quantity.	FR-17	High	1
BO-5	As a Business Owner, I would like to be able to record any order cancellations, in order to release stock and capacity.	Cancelled order is not included in the active orders anymore; its quantity and pick-up slot go back into batch.	FR-11, FR-17, FR-19	Medium	1
BO-6	As a Business Owner, I would like to be able to update the production progress in order to keep order status updated for my customers.	Status update is reflected in customer view along with notification; invalid statuses are not accepted.	FR-14, FR-15, FR-20	Medium	1
BO-7	As a Business Owner, I would like to be able to approve or deny the change requests of customers before the cut-off, in order to avoid exceeding my stock and capacity.	Order update can occur only if the stock and capacity are sufficient enough; total amount and the refunded price are updated and previous order history is preserved.	FR-22	Medium	2
BO-8	As a Business Owner, I would like to be able to have ingredient requirements calculated based on confirmed orders.	Shopping list indicates required amount of ingredients compared to currently available amount.	FR-24	Medium	2
BO-9	As a Business Owner, I would like to have ordered, cancelled, produced and collected quantities in a report so that I can plan better future batches	Report can be filtered based on batch and product and compares planned with actual amount.	FR-27	Low	3

User Story: Kitchen Staff

5. User Stories: Kitchen Staff

Kitchen Staff prepares and delivers orders. They need to be able to have a clear visibility of their tasks and to be able to track their progress and confirm delivery.

ID	User Story	Acceptance Criteria	Related FR	Priority	Sprint
KS-1	As Kitchen Staff, I need to see all the orders which are currently prepared, so I will know what should be prepared.	Order list contains the items, their variations, amount of products, and time slots for picking up of the chosen batch.	FR-18	High	1
KS-2	As Kitchen Staff, I need to see the customizations and packing details, so nothing special would be forgotten.	Customer's notes appear on each order in the production process.	FR-18	High	1
KS-3	As Kitchen Staff, I need to be able to set the order state as Preparing and Ready, so the progress could be tracked.	Only transition states of Confirmed to Preparing and Preparing to Ready are possible, and the customer receives notification about readiness of the order.	FR-14, FR-15, FR-20	High	1
KS-4	As Kitchen Staff, I need to be able to confirm the delivery of an order only in case it is Ready, so wrong handovers will not happen.	Collected state is disabled for orders which are not Ready and message appears about it.	FR-15	High	1
KS-5	As Kitchen Staff, I need cancelled orders to vanish from the production panel, so I will not prepare unnecessary orders.	Cancelled orders are invisible in the active board, but remain in the history of orders.	FR-19	Medium	1
KS-6	As Kitchen Staff, I need the ability to send the status update once even if I tapped twice, so I will not create duplicates.	Duplicate submission of the same update results in single status change and single history update.	FR-21	Medium	1
KS-7	As Kitchen Staff, I need similar products to be grouped into preparation batches, so I could work effectively.	Orders in the board are classified depending on product and variation and include notes of each customer.	FR-26	Low	2
KS-8	As Kitchen Staff, I need the ability to scan a QR code of the customer upon delivery so collection is fast and traceable.	Valid code of Ready order turns it into Collected state; invalid or used code will be rejected.	FR-25	Medium	2

Order Status Flow:
Confirmed-> Preparing->Ready->Collected. A cancelled order exits the active flow and keeps its history. No status may be skipped.

User Story: Administrator

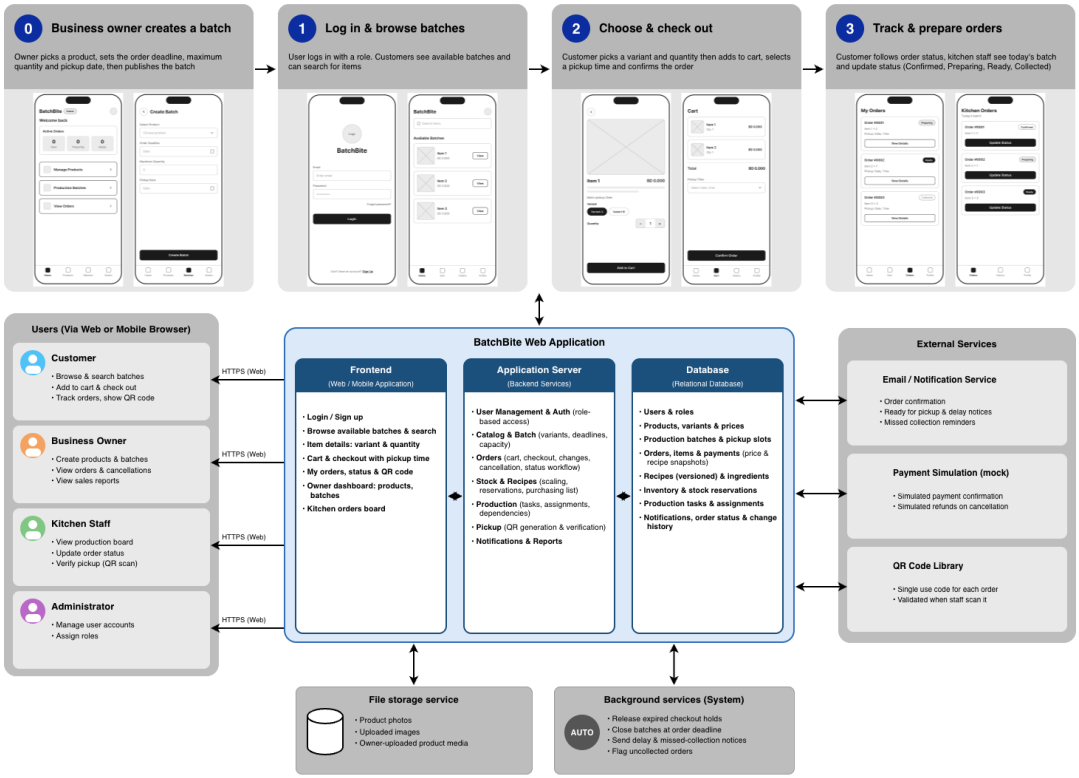
7. User Stories: Administrator

The Administrator manages user accounts and user roles to ensure that access to BatchBite is properly controlled.

ID	User Story	Acceptance Criteria	Related FR	Priority	Sprint
AD-1	As an Administrator, I would like to manage user accounts, so that I can maintain user access to BatchBite.	The Administrator can manage user accounts, and changes are reflected in the system.	FR-3	Medium	1
AD-2	As an Administrator, I would like to assign and modify user roles, so that each user has the appropriate level of access.	The Administrator can assign or modify roles for Customers, Business Owners, and Kitchen Staff. The system applies the updated permissions according to the assigned role.	FR-1, FR-2, FR-3	Medium	1

High-level Architecture Diagram

BatchBite High Level Architecture Diagram & User Flow



Wireframes

Login



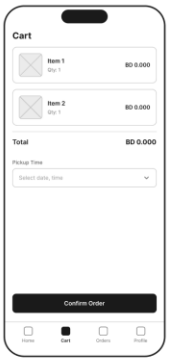
Customer Home



Product Details



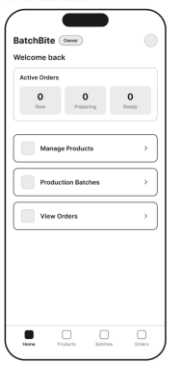
Cart & Checkout



My Orders



Owner Dashboard



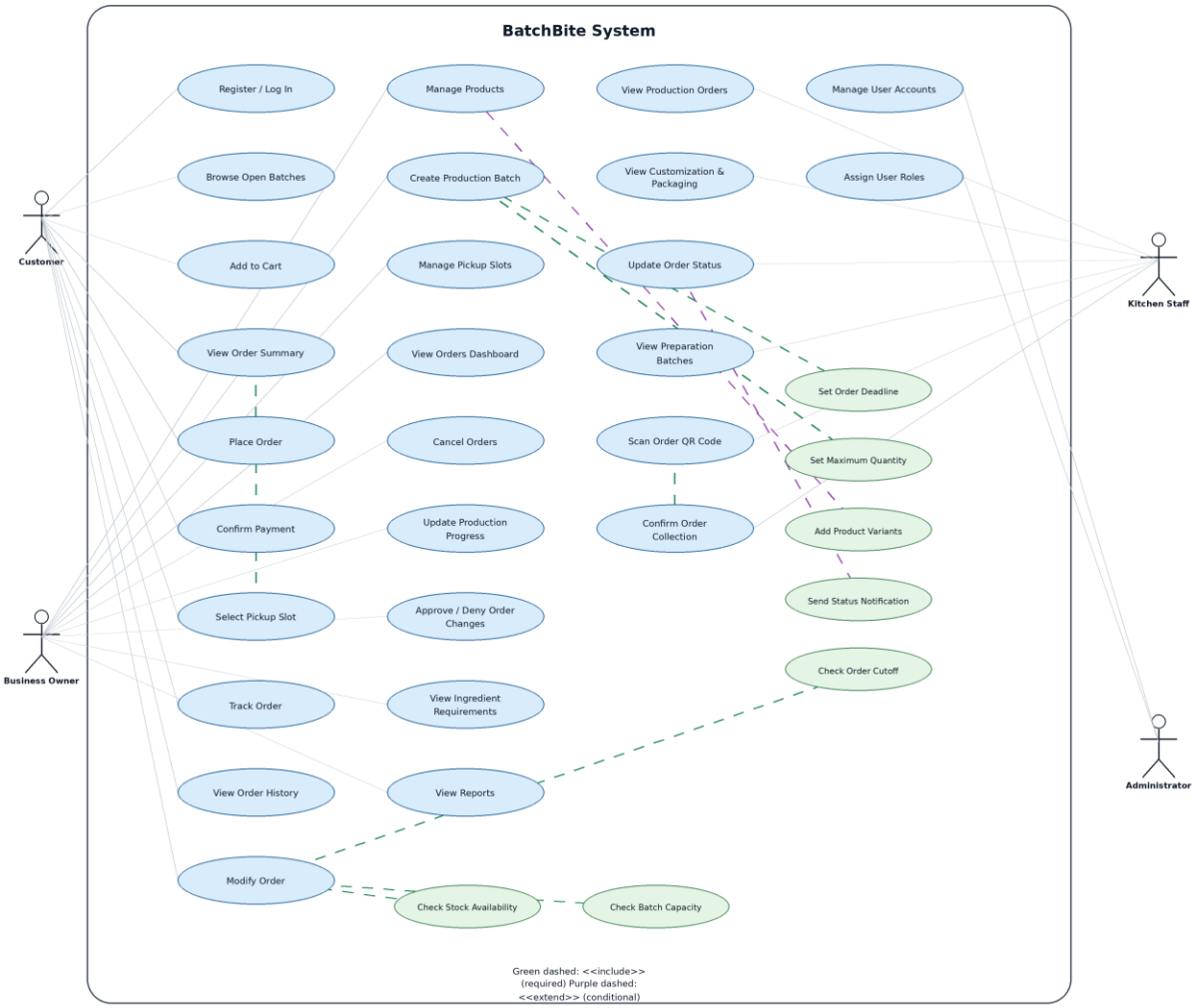
Create Production Batch



Kitchen Orders

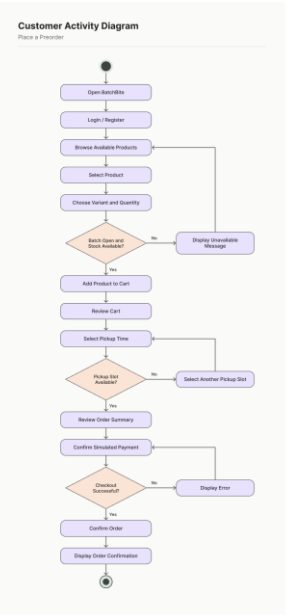


Use Case Diagram

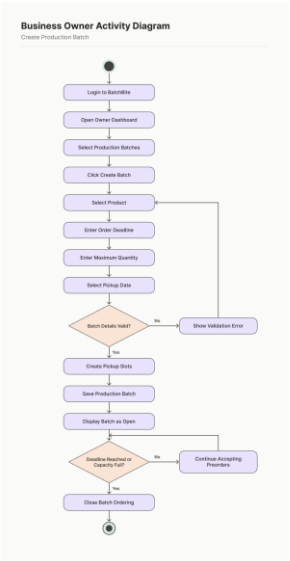


Activity Diagrams

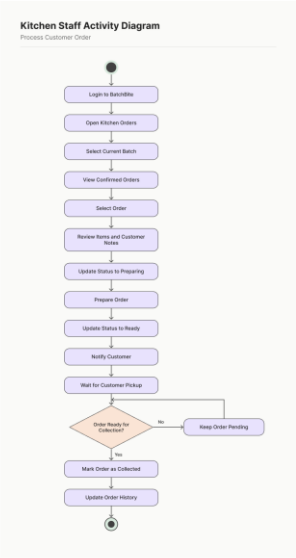
Customer Activity Diagram



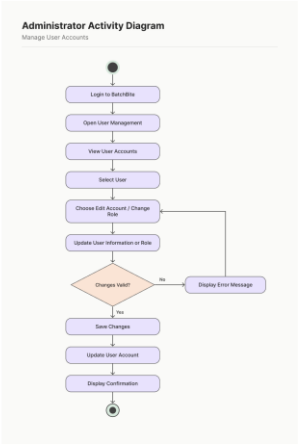
Business Owner Activity Diagram



Kitchen Staff Activity Diagram

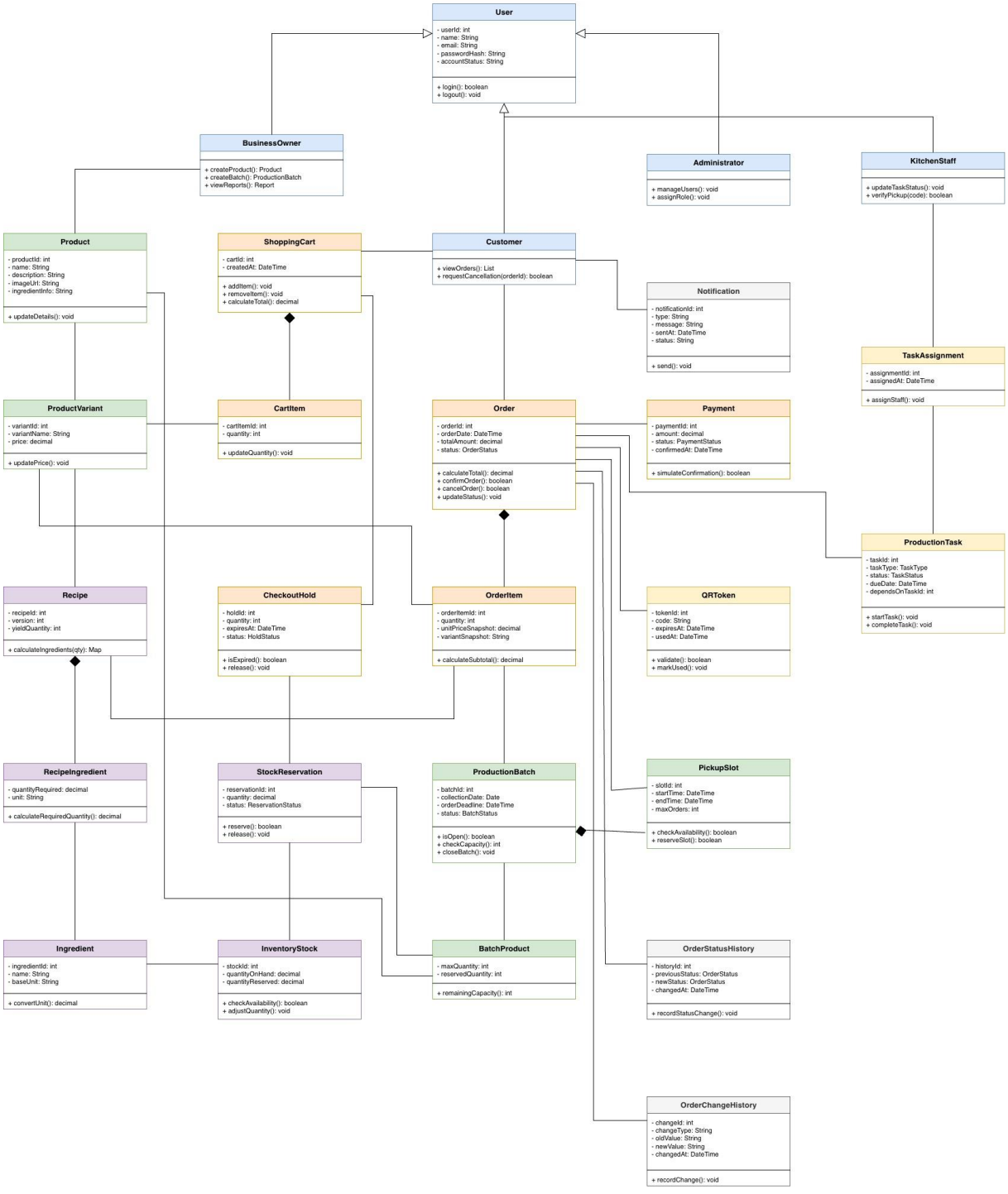


Administrator Activity Diagram



UML Class Diagram

UML Class Diagram - Batch Bites



UML Sequence Diagram

